

Data Analysis Project

GURUGRAM REAL ESTATE MARKET ANALYSIS

Turning raw data into actionable property insights

Presented by: Rohit Kumar

Objective: Analyze pricing trends and market drivers

THE PROBLEM

Why this project matters

! Messy Data

Real estate data is often scattered, inconsistent, and full of errors, making it hard to trust.

? Market Confusion

Buyers struggle to compare prices across different localities and builders without a clear guide.

THE ULTIMATE GOAL

Cleaned and analyzed 14,223 rows of real estate data to identify the key factors that drive property prices in Gurugram.

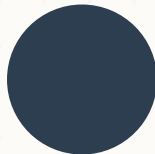
MY APPROACH

The Solution



DATA CLEANING

Standardized 12 columns (Price, Area, RERA status, etc.) to make them usable for analysis.



EDA

Used Python to analyze pricing, locality, and builder strategies to find hidden trends.



VISUALIZATION

Created clear charts to make complex numbers easy for anyone to understand.

TOOLS I USED

The Technical Stack



PYTHON

The main engine for the entire analysis, providing a flexible and powerful environment for data processing.



MATPLOTLIB & SEABORN

Used to create clear, professional, and insightful charts to communicate pricing trends to stakeholders.



PANDAS

Used for cleaning, organizing, and manipulating the 19,000+ rows of real estate data efficiently.



VS Code

For writing and testing the code step-by-step, ensuring transparency and reproducibility in the analysis.

CLEANING THE DATA

The Hardest Part of Analysis

01

REMOVING NOISE

Identified and deleted duplicate entries to ensure 14,223 unique property records.

02

STANDARDIZING UNITS

Converted prices to Crores and area to Square Feet for consistent comparison.

03

FIXING CATEGORIES

Cleaned text labels like "Ready to Move" and converted RERA status to Boolean.

LOCATION IS THE STRONGEST DRIVER

Key Finding #1

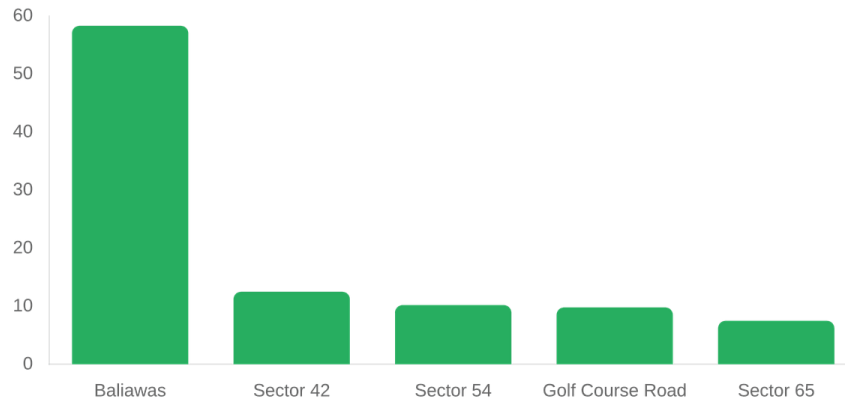
TOP LOCALITIES

Baliawas has the highest average price (58.33 Crores), while Sector 42 leads in rate per sqft (55,989).

COSTLIEST PROPERTY

An apartment in Sector 42 priced at 122.63 Crores (16,500 sqft).

Top Localities by Average Price (Crores)



Insight: Location remains the single biggest factor in determining property value in the Gurugram market.

KEY FINDING 2

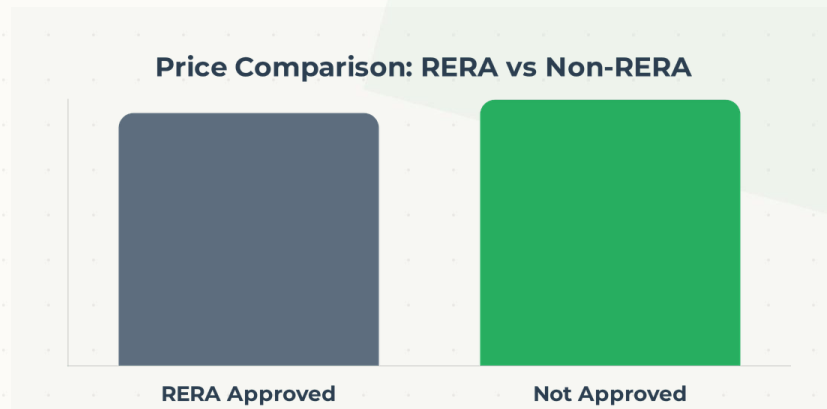
RERA Approval & Price Impact

? The Question

Do RERA-approved properties command a price premium in Gurugram?

× The Result

RERA-approved properties do NOT command a price premium in this dataset.



💡 Key Insight

Approval ensures compliance, but pricing is more influenced by location and builder brand reputation.

KEY FINDING 3

Ready-to-Move vs. Under Construction

Price Comparison by Status



Source: Gurugram Real Estate Dataset (19,000+ rows)

Ready to Move

Under Construction

THE COMPARISON

Analyzed if buyers pay more for immediate possession compared to waiting for completion.

THE RESULT

Ready-to-move properties generally cost more than those still under construction.

KEY INSIGHT

Buyers are willing to pay a premium for the convenience and reduced risk of immediate move-in.

VISUALIZING TRENDS

Area vs. Price Analysis



THE MAIN TREND

Area directly impacts total property price; larger homes have higher overall costs.

THE EXCEPTION

Luxury properties show nonlinear pricing behavior; rate per sqft doesn't always increase proportionally with area.

DATA INSIGHT

Scatter plots quickly reveal outliers in pricing vs size, especially in the premium segment.

PROPERTY TYPE & BUILDER IMPACT

Key Finding #4

PROPERTY TYPE



Villa is the most expensive flat type on average, driven by the luxury segment's premium pricing.

TOP BUILDERS (RATE/SQFT)

Camelliaaass	44,724
Cameliaas	40,000
Tulip	28,571
Prom	27,358
Magnoliaaass	26,666

Certain builders follow a premium pricing strategy, significantly impacting the rate per square foot.

PROJECT IMPACT

Real-World Results



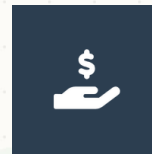
EFFICIENCY

Automated the cleaning of 14,000+ rows, saving weeks of manual work and reducing human error.



CLARITY

Provided a clear "Price Guide" for Gurugram based on actual data, not just market guesses.



DECISIONS

Helps stakeholders identify "value for money" areas and properties with high growth potential.

The project successfully turned messy real estate data into a powerful tool for informed property buying.

CONCLUSION & LEARNINGS

Final Thoughts

DATA IS POWERFUL

Even messy real estate data can tell a clear story if cleaned properly. The effort in cleaning is where the real value is found.

STORYTELLING

Learned how to translate technical Python results into simple business terms that anyone can understand and act upon.

THE NEXT STEPS

Future versions of this project could include "Price Prediction" using Machine Learning to help buyers forecast future property values.